

AutoFuzzer: Project & Firmware Specification

Embedded Communication Fuzzer & Target Firmware Robustness Testing

AutoFuzzer: Technical Challenges & Updates

1. Executive Summary & Status Updates

AutoFuzzer has successfully achieved Milestone 1 core functionality. The system currently features a 7-strategy mutation engine on ESP32, deterministic Xorshift PRNG replayability, a hardware heartbeat monitor, and a robust STM32 target parser.

2. Hardware & Soldering Challenges

Problem: OLED Display Dark on Dot Board Setup

- Symptom: OLED working 100% on female-to-female jumper wires, but dark when soldered onto dot board.

- Root Cause: Solder joints caused shorts at 7-9V) failed

3. Firmware Upload & Serial Stability

Problem: High Baud Rate Flash Failures

- Symptom: Uploads failed with 'Packet content transfer stopped (received 6 bytes)' at 921600 baud.

- Solution: S

4. Parser & Heartbeat Robustness

Problem: Target Parser Wedging & False Crash Detection

- Symptom: Truncated packets locked target parsers; CPU delays caused false crash timeouts.

- Solution: S
ESP32 heart